

One-Class Support Vector Machine

1 Motivation

In standard binary SVMs we learn a separating hyperplane between positive and negative classes. However, in many settings we have only examples from one “normal” class and wish to detect whether a new point is an outlier (“novelty detection”).

The **One-Class SVM (OC-SVM)** formalizes this by learning a decision boundary that encloses most of the training data while separating it from the origin (or equivalently, the low-density region) in feature space.

2 Problem Setup

Given training samples $\{x_i\}_{i=1}^N \subset \mathcal{X}$, we map them to a feature space $\mathcal{H}$ via $\phi : \mathcal{X} \rightarrow \mathcal{H}$ (possibly induced by a kernel). We seek a hyperplane $(\mathbf{w}, \rho)$ such that most data lie on the side $\mathbf{w}^\top \phi(x) \geq \rho$, and the hyperplane is as far as possible from the origin.

Geometric intuition. We find a small region in feature space that captures the bulk of the data with maximal margin to the origin. Points outside this region are considered anomalies.

3 Primal Optimization Problem

The OC-SVM (Schölkopf *et al.*, 2001) solves:

$$\begin{aligned} \min_{\mathbf{w}, \rho, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 - \rho + \frac{1}{\nu N} \sum_{i=1}^N \xi_i, \\ \text{s.t.} \quad & \mathbf{w}^\top \phi(x_i) \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, N. \end{aligned} \tag{1}$$

Here:

- $\mathbf{w}$ defines the normal vector of the separating hyperplane in feature space.
- ρ is the offset controlling how far the hyperplane is from the origin.
- ξ_i are slack variables allowing some data to fall below the boundary.
- $\nu \in (0, 1]$ is a parameter controlling the trade-off between outliers and margin (discussed later).

Objective interpretation.

- $\frac{1}{2}\|\mathbf{w}\|^2$: encourages a large margin.
- $-\rho$: encourages the hyperplane to be as far from the origin as possible (maximizing ρ).
- $\frac{1}{\nu N} \sum_i \xi_i$: penalizes violations (outliers).

4 Dual Derivation

We form the Lagrangian with multipliers $\alpha_i \geq 0$ for the first constraint and $\beta_i \geq 0$ for $\xi_i \geq 0$:

$$\mathcal{L}(\mathbf{w}, \rho, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{w}\|^2 - \rho + \frac{1}{\nu N} \sum_i \xi_i - \sum_i \alpha_i (\mathbf{w}^\top \phi(x_i) - \rho + \xi_i) - \sum_i \beta_i \xi_i.$$

KKT conditions. Setting partial derivatives to zero gives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 0 &\Rightarrow \mathbf{w} = \sum_i \alpha_i \phi(x_i), \\ \frac{\partial \mathcal{L}}{\partial \rho} = 0 &\Rightarrow \sum_i \alpha_i = 1, \\ \frac{\partial \mathcal{L}}{\partial \xi_i} = 0 &\Rightarrow \alpha_i + \beta_i = \frac{1}{\nu N} \Rightarrow 0 \leq \alpha_i \leq \frac{1}{\nu N}. \end{aligned}$$

Substitute into $\mathcal{L}$ to eliminate $\mathbf{w}, \rho, \boldsymbol{\xi}$:

$$\mathcal{L} = -\frac{1}{2} \sum_{i,j} \alpha_i \alpha_j \langle \phi(x_i), \phi(x_j) \rangle.$$

Hence the **dual problem** is:

$$\begin{aligned} \max_{\boldsymbol{\alpha}} \quad & -\frac{1}{2} \sum_{i,j=1}^N \alpha_i \alpha_j K(x_i, x_j), \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq \frac{1}{\nu N}, \quad \sum_{i=1}^N \alpha_i = 1, \end{aligned} \tag{2}$$

where $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$ is the kernel function.

Recovering the primal variables. After solving (2), we obtain

$$\mathbf{w} = \sum_i \alpha_i \phi(x_i), \quad \rho = \mathbf{w}^\top \phi(x_s)$$

for any *support vector* x_s satisfying $0 < \alpha_s < \frac{1}{\nu N}$.

5 Decision Function and Interpretation

A new point x is evaluated by

$$f(x) = \mathbf{w}^\top \phi(x) - \rho = \sum_i \alpha_i K(x_i, x) - \rho.$$

Decision rule:

$$\begin{cases} f(x) \geq 0, & \text{“normal” sample, inside region,} \\ f(x) < 0, & \text{“anomaly” (outside the boundary).} \end{cases}$$

Interpretation: OC-SVM builds a hyperplane separating the data from the origin. Samples with large $f(x)$ lie well inside the learned boundary; those with negative values are deemed outliers.

6 Role of the Parameter ν

The parameter $\nu \in (0, 1]$ simultaneously controls:

- An *upper bound* on the fraction of outliers (training points allowed outside the decision boundary);
- A *lower bound* on the fraction of support vectors.

Thus, ν governs the balance between the tightness of the boundary and the tolerance to anomalies:

- Small ν : conservative boundary (few points outside, wider region);
- Large ν : tighter fit, possibly overfitting with more outliers.

7 Kernelized Implementation

In practice, we use kernels such as:

$$\begin{aligned} K(x_i, x_j) &= \langle x_i, x_j \rangle && \text{(linear),} \\ K(x_i, x_j) &= \exp(-\gamma \|x_i - x_j\|^2) && \text{(RBF),} \\ K(x_i, x_j) &= (x_i^\top x_j + c)^p && \text{(polynomial).} \end{aligned}$$

The kernel implicitly defines ϕ and determines the shape of the high-dimensional boundary.

8 Connection to Density Level Sets

OC-SVM can be viewed as estimating a high-density region:

$$\mathcal{R}_\rho = \{x : \mathbf{w}^\top \phi(x) \geq \rho\}.$$

Under certain assumptions, this approximates a *level set* of the underlying density $p(x)$, i.e.

$$\mathcal{R}_\rho \approx \{x : p(x) \geq c\}.$$

Hence OC-SVM performs *nonparametric density contour estimation* using margin-based geometry.

9 Practical Considerations

- **Choice of kernel.** RBF kernels are most common; bandwidth γ affects boundary smoothness.
- **Scaling.** Features should be standardized to avoid dominance by large-scale attributes.
- **Model selection.** Tune (ν, γ) via validation on known normal and anomalous samples (if available) or heuristics such as percentile thresholds on $f(x)$.
- **Prediction scores.** $f(x)$ itself (not only its sign) can be used as an anomaly score; more negative $f(x)$ indicates stronger outlier evidence.

10 Summary Table

Concept	OC-SVM Interpretation
Training objective	Enclose data with maximal margin from origin
Constraint	$\mathbf{w}^\top \phi(x_i) \geq \rho - \xi_i$ (most data inside)
Regularization	$\frac{1}{2} \ \mathbf{w}\ ^2$ (large margin)
Trade-off parameter	ν controls outliers vs. support fraction
Decision function	$f(x) = \sum_i \alpha_i K(x_i, x) - \rho$
Outlier detection	$f(x) < 0$ indicates anomaly
Dual variables	$0 \leq \alpha_i \leq \frac{1}{\nu N}$, $\sum_i \alpha_i = 1$
Support vectors	$0 < \alpha_i < \frac{1}{\nu N}$ define boundary